

Afamelanotide (Scenesse) - Medical Clinical Policy Bulletins

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Table Of Contents

[Policy](#)

[Applicable CPT / HCPCS / ICD-10 Codes](#)

[Background](#)

[References](#)

Policy

Scope of Policy

This Clinical Policy Bulletin addresses afamelanotide (Scenesse) for commercial medical plans.

1. Criteria for Initial Approval

Aetna considers afamelanotide (Scenesse) medically necessary for the treatment of biochemically confirmed erythropoietic protoporphyria in adult members who have protoporphyrin level above the lab reference range in peripheral red blood cells.

Aetna considers all other indications as experimental, investigational, or unproven.

2. Continuation of Therapy

Aetna considers continuation of afamelanotide (Scenesse) therapy medically necessary for adult members with an indication in Section I who are experiencing benefit from therapy while receiving afamelanotide.

3. Related Policies

1. [CPB 0596 - Liver Transplantation](#)
2. [CPB 0871 - Hematopoietic Cell Transplantation for Inherited Metabolic Disorders and Genetic Diseases](#)

Dosage and Administration

Scenesse is available as an implant of 16 mg of afamelanotide.

Scenesse should be administered by a healthcare professional who is proficient in the subcutaneous implantation procedure and has completed training prior to administration.

A single implant, containing 16 mg of afamelanotide, is inserted subcutaneously above the anterior supra-iliac crest every 2 months using an SFM Implantation Cannula

Source: Clinuvel, 2024

Experimental, Investigational, or Unproven

Aetna considers afamelanotide (Scenesse) therapy experimental, investigational, or unproven for the following indications (not an all-inclusive list) because its effectiveness for these indications has not been established:

1. Inhibition of drug-resistant pathogenic bacteria (e.g., methicillin-resistant staphylococcus aureus [MRSA]);
 2. Treatment of acute ischemic stroke.
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CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
HCPCS codes covered if selection criteria is met:	
J7352	Afamelanotide implant, 1 mg
ICD-10 codes covered if selection criteria are met:	
E80.0	Hereditary erythropoietic porphyria
ICD-10 codes not covered for indications listed in the CPB (not all inclusive):	
A49.02	Methicillin resistant Staphylococcus aureus infection, unspecified site
I63.9	Cerebral infarction, unspecified

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- Scenesse is a melanocortin 1 receptor (MC1-R) agonist indicated to increase pain free light exposure in adult patients with a history of phototoxic reactions from erythropoietic protoporphyria (EPP).

Afamelanotide, branded as Scenesse (Clinuvel, Inc.), is a synthetic tridecapeptide and a structural analog of α -melanocyte stimulating hormone (α -MSH). Afamelanotide is a melanocortin receptor agonist and binds predominantly to MC1-R (Clinuvel, 2023).

The prescribing information for Scenesse includes warnings and precautions for hypersensitivity and skin monitoring. Serious hypersensitivity reactions, including anaphylaxis, have been reported with postmarket use. Moreover, afamelanotide may induce darkening of pre-existing nevi and ephelides due to its pharmacological effect. A regular full body skin examination (twice yearly) is recommended to monitor all nevi and other skin abnormalities.

The most common adverse reactions (incidence greater than 2 percent) include implant site reaction, nausea, oropharyngeal pain, cough, fatigue, dizziness, skin hyperpigmentation, somnolence, melanocytic nevus, respiratory tract infection, non-acute porphyria, and skin irritation.

Erythropoietic Protoporphyria

The term "protoporphyria" includes both erythropoietic protoporphyria (EPP) and X-linked protoporphyria (XLP), genetic disorders that cause a buildup of protoporphyrin in the body, leading to sensitivity to sunlight. EPP is caused by mutations in the FECH gene,

while XLP is caused by mutations in the ALAS2 gene located on the X chromosome. This leads to differences in the severity of symptoms and inheritance patterns (United Porphyrias Association, 2024).

EPP is an autosomal recessive inherited cutaneous porphyria characterized by severe burning pain in the skin within minutes of sun exposure, but leaving little residual skin damage. This photosensitivity is usually first noted in early childhood and may be misdiagnosed as an allergic reaction or primary angioedema. The predominant clinical manifestation in EPP and XLP is painful, non-blistering cutaneous photosensitivity that differs distinctly from the chronic, blistering skin manifestations of the other cutaneous porphyrias. Hepatobiliary complications include protoporphyrin-containing gallstones and, in less than 5 percent of cases, severe liver failure. The cutaneous phenotype can result from altered activity of one of two enzymes in the heme biosynthetic pathway, either a deficiency of ferrochelatase (FECH), which causes EPP; or a gain-of-function mutation of the erythroid-specific form of delta-aminolevulinic acid synthase (ALAS2), which causes XLP. Impairment of the FECH enzyme results in the build-up of protoporphyrin in the bone marrow, red blood cells, blood plasma, skin, and eventually liver. Build up of protoporphyrin can cause extreme sensitivity to sunlight, liver damage, abdominal pain, gallstones, and enlargement of the spleen (Poh-Fitzpatrick, 2016). An acquired, adult-onset form of EPP has also been described, in which a clone of cells with mutated FECH expands in the setting of a myeloproliferative or myelodysplastic syndrome. All forms of porphyria afflict fewer than 200,000 people in the United States (American Porphyrria Association).

Although the upper limit of normal can vary with age and among laboratories, the values in patients with EPP are markedly elevated, to approximately 300 to 8,000 mcg/dL. If total erythrocyte protoporphyrin is elevated, the total should be fractionated, and so that the relative proportion (or amounts) of zinc protoporphyrin versus metal-free protoporphyrin can be derived. In EPP and most cases of XLP, the excess is predominantly metal-free protoporphyrin, whereas in other conditions, it is mostly zinc protoporphyrin. The defining laboratory manifestation of EPP is a marked elevation of total erythrocyte protoporphyrin that is mostly metal-free (85 to 100 % metal-free protoporphyrin); the defining finding in XLP is marked elevation of total erythrocyte protoporphyrin that is approximately 50 to 85 % metal-free protoporphyrin (Mittal, 2019).

Historically, the primary measures to manage EPP has been to avoid sunlight or fluorescent light, which can greatly impair daily activities and quality of life. Other forms of photoprotection and other interventions include protective clothing, hats, and protective tinted automobile window glass. Several small studies and case series also reported increased tolerance to sunlight with beta-carotene, especially in the summer. The proposed protective mechanism is quenching of oxygen free radicals by beta-carotene (Mittal, 2019).

On October 08, 2019, the FDA approved afamelanotide (Scenesse) to increase pain free light exposure in adult patients with a history of phototoxic reactions from erythropoietic protoporphyria (EPP). Afamelanotide is a synthetic analogue of alpha-melanocyte stimulating hormone (alpha-MSH), a naturally occurring hormone that increases skin pigmentation by increasing melanin production, and reduces free radical formation and cytokine production. Afamelanotide, the first medication approved for the treatment of EPP, was granted priority review and orphan drug designation by the FDA and was approved for use in Europe in 2014 by the European Medicines Agency. The FDA approval of Scenesse was based upon the results from 3 vehicle-controlled, parallel-group clinical in subjects with EPP.

In 2010, Clinuvel completed its first phase-III study of afamelanotide in patients with EPP (CUV017). A total of 91 patients completed the 12-month study, in which an 11-point Likert scale and physician assessments through case report forms (CRF) were used to evaluate pain as a principal symptom of phototoxicity. The duration of daily (sun)light exposure was used to assess the willingness of patients to expose themselves during all seasons. Melanin density (reflecting changes in skin pigmentation, measured by spectrophotometry) and quality of life (Short Form 36 surveys) were also evaluated. In an analysis of the total number of days (frequency distribution) on which patients experienced pain in the specific pain severity categories (severe, moderate, mild and none), a significant reduction of frequency was observed in patients on active drug [$p = 0.0023$]. Characteristic to EPP, the majority of phototoxic reactions occurred during spring and summer. In analyzing the average pain severity experienced by the total number of patients, the assessment of all individual daily pain scores was significantly lower in patients receiving afamelanotide compared to those receiving placebo [$p = 0.0017$]. An

additional evaluation of the pain scores in patients willing to modify behavior by continuous exposure to daily (sun)light showed a positive trend toward a reduction in average pain score following active drug treatment [$p = 0.1654$] (Langendonk, 2015).

Two trials (U.S. study CUV039, NCT 01605136; and the EU Study CUV029, NCT 00979745) were designed to assess exposure to direct sunlight on days with no phototoxic pain. The 2 trials differed in the number of days of follow-up, the time windows within a day in which time spent outdoors was recorded, and how the amount of time spent in direct sunlight on each day was characterized. The subjects enrolled in these trials were primarily Caucasian (98 %), the mean age was 40 years (range of 18 to 74 years), and 53 % of subjects were male and 47 % were female (Langendonk, 2015).

Study CUV039 enrolled 93 subjects, of whom 48 received 16 mg of afamelanotide administered subcutaneously every 2 months and 45 received vehicle. Subjects received three implants and were followed for 180 days. On each study day, subjects recorded the number of hours spent in direct sunlight between 10 am and 6 pm, the number of hours spent in shade between 10 am and 6 pm, and whether they experienced any phototoxic pain that day. The primary endpoint was the total number of hours over 180 days spent in direct sunlight between 10 am and 6 pm on days with no pain. The median total number of hours over 180 days spent in direct sunlight between 10 am and 6 pm on days with no pain was 64.1 hours for subjects receiving afamelanotide and 40.5 hours for subjects receiving vehicle.

Study CUV029 enrolled 74 subjects, of whom 38 received afamelanotide (16 mg of afamelanotide administered subcutaneously every 2 months), 36 received vehicle. Subjects received five implants and were followed for 270 days. On each study day, subjects recorded the number of hours spent outdoors between 10 am and 3 pm, whether “most of the day” was spent in direct sunlight, shade, or a combination of both, and whether they experienced any phototoxic pain that day. The primary endpoint was the total number of hours over 270 days spent outdoors between 10 am and 3 pm on days with no pain for which “most of the day” was spent in direct sunlight. This analysis does not include sun exposure on days for which subjects reported spending time in a combination of both direct sunlight and shade. The median total number of hours over 270 days spent outdoors between 10 am and 3 pm on days with no pain for which “most of the day” was spent in direct sunlight was 6.0 hours for subjects in the afamelanotide group and 0.75 hours for subjects in the vehicle group.

Stolzel et al (2019) stated that physicians should be aware of porphyrias, which could be responsible for unexplained gastrointestinal, neurologic, or skin disorders. Despite their relative rarity and complexity, most porphyrias can be easily defined and diagnosed. They are caused by well-characterized enzyme defects in the complex heme biosynthetic pathway and are divided into categories of acute vs non-acute or hepatic vs erythropoietic porphyrias. Acute hepatic porphyrias (acute intermittent porphyria, variegate porphyria, hereditary coproporphyria, and aminolevulinic acid dehydratase deficient porphyria) manifest in attacks and are characterized by overproduction of porphyrin precursors, producing often serious abdominal, psychiatric, neurologic, or cardiovascular symptoms. Patients with variegate porphyria and hereditary coproporphyria can present with skin photosensitivity. Diagnosis relies on measurement of increased urinary 5-aminolevulinic acid (in patients with aminolevulinic acid dehydratase deficient porphyria) or increased 5-aminolevulinic acid and porphobilinogen (in patients with other acute porphyrias). Management of attacks requires intensive care, strict avoidance of porphyrinogenic drugs and other precipitating factors, caloric support, and often heme therapy. The non-acute porphyrias are porphyria cutanea tarda, erythropoietic protoporphyria, X-linked protoporphyria, and the rare congenital erythropoietic porphyria. They lead to the accumulation of porphyrins that cause skin photosensitivity and occasionally severe liver damage. Secondary elevated urinary or blood porphyrins can occur in patients without porphyria, for example, in liver diseases, or iron deficiency. Increases in porphyrin precursors and porphyrins are also found in patients with lead intoxication. Patients with porphyria cutanea tarda benefit from iron depletion, hydroxychloroquine therapy, and, if applicable, elimination of the hepatitis C virus. An α -melanocyte-stimulating hormone analogue can reduce sunlight sensitivity in patients with erythropoietic protoporphyria or X-linked protoporphyria. Strategies to address dysregulated or dysfunctional steps within the heme biosynthetic pathway are in development.

Minder et al (2017) stated that afamelanotide, the first α -melanocyte-stimulating hormone (MSH) analogue, synthesized in 1980, was broadly investigated in all aspects of pigmentation because its activity and stability were higher than the natural hormone.

Afamelanotide binds to the melanocortin-1 receptor (MC1R), and MC1R signaling increases melanin synthesis, induces antioxidant activities, enhances DNA repair processes and modulates inflammation. The loss-of-function variants of the MC1R present in fair-skinned Caucasians are less effectively activated by the natural hormone. Afamelanotide was the first α -MSH analogue to be applied to human volunteers. Ten daily doses of between 0.08 and 0.21 mg/kg in saline injected subcutaneously resulted in long-lasting skin pigmentation and enabled basic pharmacokinetics. Subcutaneous application had full bioavailability, but neither oral nor transdermal application resulted in measurable plasma concentrations or pigmentation response. Two trials in human volunteers showed that neither MC1R variants nor fair skin reduced the afamelanotide-induced increase in skin pigmentation. A controlled-release formulation optimizes administration in man and is effective at a lower dose than the daily saline injections. Promising therapeutic results were published in polymorphic light eruption, EPP, solar urticaria, Hailey-Hailey disease and vitiligo. In 2014, afamelanotide was approved by the European Medicines Agency for the prevention of phototoxicity in adult patients with EPP. No late effects were reported in volunteers 25 years after the first exposure or after continuous long-term application of up to 8 years in EPP patients, and an immunogenic potential has been excluded. Generally, adverse effects were benign in all trials.

Other Indications

Inhibition of Drug-Resistant Pathogenic Bacteria (e.g., Methicillin-Resistant *Staphylococcus Aureus* [MRSA])

Otarigho and Falade (2023) noted that antibiotic resistance is a serious problem that results in a high morbidity and mortality rate. The process of discovering new chemotherapy and antibiotics is challenging, expensive, and time-consuming, with only a few getting approved for clinical use; thus, screening already-approved drugs to combat pathogens such as bacteria that cause serious infections in humans and animals is highly encouraged. These researchers identified approved antibiotics that can inhibit the *mecA* antibiotic resistance gene found in methicillin-resistant *Staphylococcus aureus* (MRSA) strains. The *MecA* protein sequence was employed to carry out a BLAST search against a drug database containing 4,302 approved drugs. The results revealed that 50 medications, including known antibiotics for other bacterial strains, targeted the *mecA* antibiotic resistance gene. Furthermore, a structural similarity approach was used to identify existing antibiotics for *S. aureus*, followed by molecular docking. The results of the docking experiment indicated that 6 drugs had a high binding affinity to the *mecA* antibiotic resistance gene. In addition, using the structural similarity strategy, it was found that afamelanotide, an FDA-approved drug with unclear antibiotic activity, exhibited a strong binding affinity to the MRSA-MecA protein. The authors stated that these findings suggested that certain already-approved drugs have potential in chemotherapy against drug-resistant pathogenic bacteria, such as MRSA.

Melanocytic Nevi

Arisi and colleagues (2021) noted that afamelanotide (AFA) is a synthetic analogue of α -melanocyte-stimulating hormone that is FDA-approved for the treatment of patients with EPP. AFA induces a "sun free" tanning and changes of acquired melanocytic nevi (AMN) that are generically described as "darkening". These researchers examined clinical and dermoscopic AMN changes during AFA treatment. Adult EPP patients treated with 2 AFA implants 50 days apart were enrolled. They underwent a clinical and dermoscopic examination of all AMN at baseline (T0), and after 5 (T1) and 12 (T2) months from the 1st AFA implant. The general pattern, symmetry, number, and size of pigmented globules, morphology of the pigment network, and dermoscopic melanoma features were evaluated. A total of 15 patients were enrolled with 103 AMN. At T1 all reticular and 2-component AMN showed a focal network thickening that returned to baseline by T2. The increase of globules' number was observed at T1 but not at T2. The difference in number was not influenced by patients' age or phototype. Dermoscopic changes suggestive of malignancy were never observed. The development of new AMN was never registered. The authors concluded that AFA-induced MC1R stimulation of AMN melanocytes appeared to induce only transient physiological changes of AMN morphology and as such, they do not support the concerns of carcinogenic activity of AFA. However, the role of AFA in promoting the growth of a melanoma already present remains unaddressed, a careful dermoscopic evaluation before treatment with AFA should be mandatory and long-term safety of AFA treatment has to be further examined.

The authors stated that the findings of this study could be affected by the limited number of patients (n = 15) enrolled; however, they could also support the ability of AFA to induce AMN changes independently from constitutive pigmentogenic activity. Another drawback of this study was not to have ascertained AMN modifications by spectrophotometric evaluation or histology. Moreover, this study did not examine the effects of AFA on a pre-existing melanoma or pre-cancerous lesion.

Treatment of Acute Ischemic Stroke

Stanislaus et al (2023) stated that neuroprotective agents have the potential to improve the outcomes of re-vascularization therapies in acute ischemic stroke patients (AIS) and in those unable to receive re-vascularization. Afamelanotide is a potential novel neuroprotective agent. In an open-label, proof-of-concept (POC), phase-IIa clinical trial, these researchers examined the safety and feasibility of afamelanotide in AIS patients. Subjects were AIS patients within 24 hours of onset, with perfusion abnormality on imaging (Tmax) and otherwise ineligible for re-vascularization therapies.

Afamelanotide implants (16 mg) were administered subcutaneously on Day 0 (D0, day of recruitment), D1 and repeated on D7 and D8, if not well-recovered. Treatment emergent adverse events (TEAEs) and neurological assessments were recorded regularly up to D42.

Magnetic resonance imaging (MRI) with FLAIR sequences were also performed on D3 and D9. A total of 6 patients (5 women, median age of 81 years, median National Institutes of Health Stroke Scale [NIHSS] of 6) were recruited – 2 patients received 4 doses and 4 patients received 2. One patient (who received 2 doses), suffered a fatal recurrent stroke on D9 due to a known complete acute internal carotid artery occlusion, assessed as unrelated to the study drug. There were no other local or major systemic TEAEs recorded.

In all surviving patients, the median NIHSS improved from 6 to 2 on D7. The median Tmax volume on D0 was 23 ml, which was reduced to a FLAIR volume of 10 ml on D3 and 4 ml on D9. The authors concluded that afamelanotide was safe and well-tolerated in a small sample of AIS patients. It also appeared to be associated with good recovery and radiological improvement of salvageable tissue. Moreover, these researchers stated that these findings need to be tested in large, randomized studies.

The authors stated that drawbacks this study included a small study population (n = 6) with gender disproportion (5 women and 1 man). Although no serious AEs were observed in this phase-IIa feasibility study, a larger sample size is needed to draw reliable conclusions regarding afamelanotide's safety and effectiveness. All participants had NIHSS of less than 9, suggesting mild-to-moderate stroke. It could be argued that the changes observed in NIHSS with time was likely due to the natural recovery from stroke; however, this may not explain the radiological changes and randomized placebo-controlled trials are needed.

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